

Research Notes on Bayesian network

Bayesian network

$$P(h | D) = \frac{P(D | h)P(h)}{P(D)}$$

>> Section 1

A back-door path is one that ends with an arrow into X . Sets that satisfy the back-door criterion are called "sufficient" or "admissible." For example, the set $Z = R$ is admissible for predicting the effect of $S = T$ on G , because R d-separates the (only) back-door path $S \leftarrow R \rightarrow G$. However, if S is not observed, no other set d-separates this path and the effect of turning the sprinkler on ($S = T$) on the grass (G) cannot be predicted from passive observations. In that case $P(G | \text{do}(S = T))$ is not "identified". This reflects the fact that, lacking interventional data, the observed dependence between S and G is due to a causal connection or is spurious (apparent dependence arising from a common cause, R). (see Simpson's paradox) To determine whether a causal relation is identified from an arbitrary Bayesian network with unobserved variables, one can use the three rules of "do-calculus" and test whether all do terms can be removed from the expression of that relation, thus confirming that the desired quantity is estimable from frequency data. Using a Bayesian network can save considerable amounts of memory over exhaustive probability tables, if the dependencies in the joint distribution are sparse. For example, a naive way of storing the conditional probabilities of 10 two-valued variables as a table requires storage space for $2^{10} = 1024$ values. If no variable's local distribution depends on more than three parent variables, the Bayesian network representation stores at most $10 \cdot 2^3 = 80$ values. One advantage of Bayesian networks is that it is intuitively easier for a human to understand (a sparse set of) direct dependencies and local distributions than complete joint distributions.

>> Section 2

obtained by removing the factor $\Pr(G \mid S, R)$ from the pre-intervention distribution. The do operator forces the value of G to be true. The probability of rain is unaffected by the action:

>> Section 3

Graphical model Formally, Bayesian networks are directed acyclic graphs (DAGs) whose nodes represent variables in the Bayesian sense: they may be observable quantities, latent variables, unknown parameters or hypotheses. Each edge represents a direct conditional dependency. Any pair

of nodes that are not connected (i.e. no path connects one node to the other) represent variables that are conditionally independent of each other. Each node is associated with a probability function that takes, as input, a particular set of values for the node's parent variables, and gives (as output) the probability (or probability distribution, if applicable) of the variable represented by the node. For example, if m parent nodes represent m Boolean variables, then the probability function could be represented by a table of 2^m entries, one entry for each of the 2^m possible parent combinations. Similar ideas may be applied to undirected, and possibly cyclic, graphs such as Markov networks.

>> Section 4

Local Markov property X is a Bayesian network with respect to G if it satisfies the local Markov property: each variable is conditionally independent of its non-descendants given its parent variables:

>> Section 5

Structure learning In the simplest case, a Bayesian network is specified by an expert and is then used to perform inference. In other applications, the task of defining the network is too complex for humans. In this case, the network structure and the parameters of the local distributions must be learned from data. Automatically learning the graph structure of a Bayesian network (BN) is a challenge pursued within machine learning. The basic idea goes back to a recovery algorithm developed by Rebane and Pearl and rests on the distinction between the three possible patterns allowed in a 3-node DAG:

>> Section 6

To answer an interventional question, such as "What is the probability that it would rain, given that we wet the grass?" the answer is governed by the post-intervention joint distribution function

>> Section 7

where G = "Grass wet (true/false)", S = "Sprinkler turned on (true/false)", and R = "Raining (true/false)". The model can answer questions about the presence of a cause given the presence of an effect (so-called inverse probability) like "What is the probability that it is raining, given the grass is wet?" by using the conditional probability formula and summing over all nuisance variables:

>> Section 8

d-separation This definition can be made more general by defining the "d"-separation of two nodes, where d stands for directional. We first define the "d"-separation of a trail

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and then we will define the "d"-separation of two nodes in terms of that. Let P be a trail from node u to v . A trail is a loop-free, undirected (i.e. all edge directions are ignored) path between two nodes. Then P is said to be d-separated by a set of nodes Z if any of the following conditions holds:

>> Section 9

$$\Pr(G, S, R) = \Pr(G \mid S, R) \Pr(S \mid R) \Pr(R)$$
$$\{\displaystyle \Pr(G,S,R)=\Pr(G\mid S,R)\Pr(S\mid R)\Pr(R)\}$$

>> Section 10

Restrictions on priors Some care is needed when choosing priors in a hierarchical model, particularly on scale variables at higher levels of the hierarchy such as the variable τ $\{\displaystyle \tau \}$ in the example. The usual priors such as the Jeffreys prior often do not work, because the posterior distribution will not be normalizable and estimates made by minimizing the expected loss will be inadmissible.

>> Section 11

Several equivalent definitions of a Bayesian network have been offered. For the following, let $G = (V, E)$ be a directed acyclic graph (DAG) and let $X = (X_v)$, $v \in V$ be a set of random variables indexed by V .

>> Section 12

Using the expansion for the joint probability function $\Pr(G, S, R)$ $\{\displaystyle \Pr(G,S,R)\}$ and the conditional probabilities from the conditional probability tables (CPTs) stated in the diagram, one can evaluate each term in the sums in the numerator and denominator. For example,

>> Section 13

A Bayesian network (also known as a Bayes network, Bayes net, belief network, or decision network) is a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG). While it is one of several forms of causal notation, causal networks are special cases of Bayesian networks. Bayesian networks are ideal for taking an event that occurred and predicting the likelihood that any one of several possible known causes was the contributing factor. For example, a Bayesian network could represent the probabilistic relationships between diseases and symptoms. Given symptoms, the network can be used to compute the probabilities of the presence of various diseases. Efficient algorithms can perform inference and learning in Bayesian networks. Bayesian networks that model sequences of variables (e.g. speech signals or protein sequences) are called dynamic Bayesian networks. Generalizations of Bayesian networks that can represent

and solve decision problems under uncertainty are called influence diagrams.